SDS 4030 - Statistics for Data Science II

Final Project Code

Problem 1. *Proof.*

```r
# loading dataset Here

horse = read.table("horse.txt", header = TRUE)
head(horse)
```

```
##   surgery   age temperature pulse respiratory_rate temperature_of_extremities
## 1       0 adult        38.5    66               28                       cool
## 2       0 adult        38.3    40               24                     normal
## 3       1 young        39.1   164               84                       cold
## 4       1 adult        37.9    48               16                     normal
## 5       0 adult        38.6    80               36                       cool
## 6       1 adult        38.1    66               12                       cool
##   peripheral_pulse       pain peristalsis abdominal_distension
## 1          reduced continuous      absent               severe
## 2           normal       mild  hypomotile                 none
## 3           normal  depressed      absent               severe
## 4           normal       mild  hypomotile             moderate
## 5           absent     severe      absent               severe
## 6          reduced       mild  hypomotile                 none
##   packed_cell_volume total_protein    outcome surgical_lesion
## 1                 45           8.4       died               0
## 2                 33           6.7      lived               0
## 3                 48           7.2       died               1
## 4                 37           7.0      lived               1
## 5                 38           6.2 euthanized               1
## 6                 44           6.0      lived               1
```

```r
# starter EDA on resp rate
library(ggplot2)
```

```
## Warning: package 'ggplot2' was built under R version 4.5.2
```

```r
resp_summary <- summary(horse$respiratory_rate)
resp_summary
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
```

```
##    2.00   17.75   28.00   29.96   36.00   90.00
```

```
mean_resp <- mean(horse$respiratory_rate, na.rm = TRUE)
median_resp <- median(horse$respiratory_rate, na.rm = TRUE)
min_resp <- min(horse$respiratory_rate, na.rm = TRUE)
max_resp <- max(horse$respiratory_rate, na.rm = TRUE)

cat("Mean:", mean_resp, "\n")
```

```
## Mean: 29.95833
```

```
cat("Median:", median_resp, "\n")
```

```
## Median: 28
```

```
cat("Min:", min_resp, "\n")
```

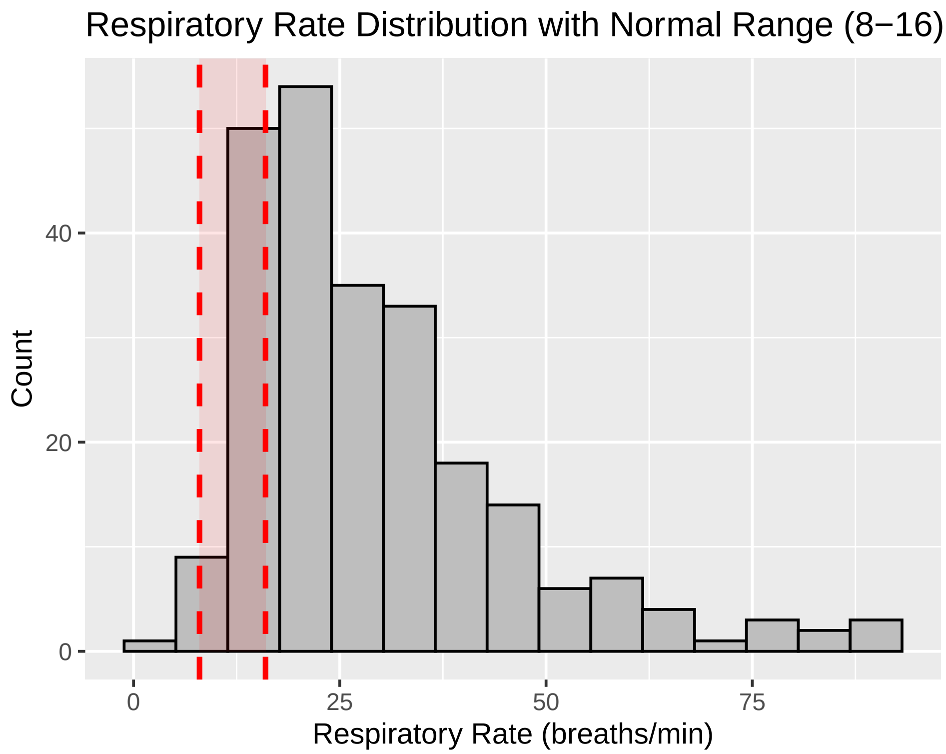
```
## Min: 2
```

```
cat("Max:", max_resp, "\n")
```

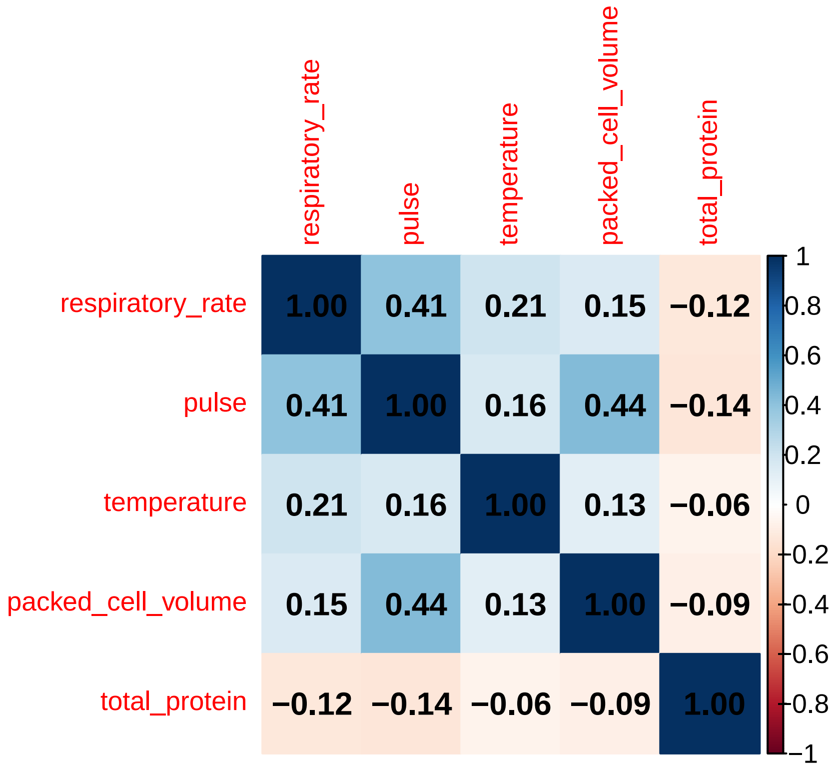
```
## Max: 90
```

```
ggplot(horse, aes(x = respiratory_rate)) + geom_histogram(bins = 15,
    fill = "gray", color = "black") + geom_vline(xintercept = 8, linetype = "dashed",
    color = "red", size = 1) + geom_vline(xintercept = 16, linetype = "dashed",
    color = "red", size = 1) + annotate("rect", xmin = 8, xmax = 16,
    ymin = 0, ymax = Inf, alpha = 0.1, fill = "red") + labs(title = "Respiratory Rate Di
    x = "Respiratory Rate (breaths/min)", y = "Count")
```

```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```




```r
# correlation matrix for now

library(corrplot)

numeric_vars <- c("respiratory_rate", "pulse", "temperature", "packed_cell_volume",
    "total_protein")
corr_matrix <- cor(horse[, numeric_vars], use = "complete.obs")

corrplot(corr_matrix, method = "color", addCoef.col = "black", tl.cex = 0.8)
```




```
# fitted model w only quant variables

quant_model = lm(respiratory_rate ~ temperature + pulse + packed_cell_volume +
    total_protein, data = horse)
summary(quant_model)
```

```
##
## Call:
## lm(formula = respiratory_rate ~ temperature + pulse + packed_cell_volume +
##     total_protein, data = horse)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -32.852 -10.009  -3.102   7.040  56.187
##
## Coefficients:
##                     Estimate Std. Error t value Pr(>|t|)
## (Intercept)       -112.98010   52.66873  -2.145   0.0330 *
## temperature          3.39913    1.39221   2.442   0.0154 *
## pulse                0.25250    0.04231   5.967 8.82e-09 ***
## packed_cell_volume  -0.06932    0.10832  -0.640   0.5228
## total_protein       -0.03715    0.03676  -1.011   0.3132
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

```
## Residual standard error: 15.32 on 235 degrees of freedom
## Multiple R-squared:  0.1916, Adjusted R-squared:  0.1779
## F-statistic: 13.93 on 4 and 235 DF,  p-value: 3.282e-10
```

```r
# qual, exact, large sample size -- lecture 21 reference


beta <- coef(quant_model)
X <- model.matrix(quant_model)
Y = horse$respiratory_rate
n <- nrow(X)
p <- ncol(X)
S2 <- sum(residuals(quant_model)^2)/(n - p)


R <- matrix(c(0, 5, 0, 0, 350, 0, 0, 40, 100, 0), nrow = 2, byrow = TRUE)

r <- c(3, 3)
k <- length(r)

FT <- crossprod(R %*% beta - r, solve(R %*% solve(t(X) %*% X) %*% t(R),
    R %*% beta - r))/(k * S2)
print(FT)
```

```
##             [,1]
## [1,] 0.002356961
```

```r
pvalue <- pf(FT, k, n - p, lower.tail = FALSE)
print(pvalue)
```

```
##           [,1]
## [1,] 0.9976458
```

```r
quant_reduced <- lm(respiratory_rate ~ I(temperature + total_protein +
    pulse + packed_cell_volume), data = horse)
summary(quant_reduced)
```

```
##
## Call:
## lm(formula = respiratory_rate ~ I(temperature + total_protein +
##     pulse + packed_cell_volume), data = horse)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -28.046 -12.065  -3.351   6.948  62.740
##
## Coefficients:
```

```
##                                                        Estimate Std. Error
## (Intercept)                                            12.21057    4.89454
## I(temperature + total_protein + pulse + packed_cell_volume)  0.10100    0.02719
##                                                        t value Pr(>|t|)
## (Intercept)                                              2.495 0.013284
## I(temperature + total_protein + pulse + packed_cell_volume)   3.715 0.000254
##
## (Intercept)                                            *
## I(temperature + total_protein + pulse + packed_cell_volume) ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 16.47 on 238 degrees of freedom
## Multiple R-squared:  0.0548, Adjusted R-squared:  0.05083
## F-statistic:  13.8 on 1 and 238 DF,  p-value: 0.0002536
```

```
anova(quant_reduced, quant_model)
```

```
## Analysis of Variance Table
##
## Model 1: respiratory_rate ~ I(temperature + total_protein + pulse + packed_cell_volum
## Model 2: respiratory_rate ~ temperature + pulse + packed_cell_volume +
##     total_protein
##   Res.Df   RSS Df Sum of Sq      F    Pr(>F)
## 1    238 64523
## 2    235 55183  3    9339.9 13.258 5.016e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# qual, asymptotic, large sample size -- lecture 21 reference

WT = k * FT
print(WT)
```

```
##             [,1]
## [1,] 0.004713923
```

```
pvalue = pchisq(WT, k, lower.tail = FALSE)
print(pvalue)
```

```
##           [,1]
## [1,] 0.9976458
```

```
anova(quant_reduced, quant_model, test = "Chisq")
```

```
## Analysis of Variance Table
##
## Model 1: respiratory_rate ~ I(temperature + total_protein + pulse + packed_cell_volum
```

```
## Model 2: respiratory_rate ~ temperature + pulse + packed_cell_volume +
##     total_protein
##   Res.Df   RSS Df Sum of Sq  Pr(>Chi)
## 1    238 64523
## 2    235 55183  3    9339.9 1.189e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The conclusions agree of both the exact and asymptotic tests agree with each other. Both had a p-value of 0.9043546, so we fail to reject the null hypothesis. The Wald statistic and Exact F statistic are related as

$$W_n = \frac{\sigma}{S_{nQ1}^2} = kF$$

where F is the F statistic from the exact test. So we would expect the conclusions of both tests to agree with each other.

□

Problem 2. *Proof.*

```r
# jackknife
betajack <- matrix(0, n, p)
for (i in 1:n) {
    jack <- lm(respiratory_rate ~ temperature + pulse + packed_cell_volume +
        total_protein, data = horse[-i, ])
    betajack[i, ] <- coef(jack)
}
betabar <- colMeans(betajack)
print(betabar)
```

```
## [1] -112.97355280    3.39896234    0.25248853   -0.06931150   -0.03715342
```

```r
SE <- apply(betajack, 2, sd) * sqrt((n - 1)^2/n)
print(SE)
```

```
## [1] 57.37518872  1.53850267  0.04898266  0.12188824  0.03813573
```

```r
# nonparametric bootstrap
nboot <- 10000
betaboot <- matrix(0, nboot, p)
set.seed(123)

for (k in 1:nboot) {
    indices <- sample(1:n, n, replace = TRUE)
    boot <- lm(respiratory_rate ~ temperature + pulse + packed_cell_volume +
        total_protein, data = horse[indices, ])
```

```r
    betaboot[k, ] <- coef(boot)
}

betabar <- colMeans(betaboot)
print(betabar)
```

```
## [1] -110.53315091    3.33575200    0.25106308   -0.06812419   -0.03683839
```

```r
SE <- apply(betaboot, 2, sd)
print(SE)
```

```
## [1] 57.58995133  1.54368567  0.04753332  0.11860789  0.03728970
```

```r
alpha <- 0.05

CInormal <- cbind(betabar - qnorm(1 - alpha/2) * SE, betabar + qnorm(1 -
    alpha/2) * SE)
print(CInormal)
```

```
##              [,1]       [,2]
## [1,] -223.4073814 2.34107957
## [2,]    0.3101837 6.36132031
## [3,]    0.1578995 0.34422667
## [4,]   -0.3005914 0.16434300
## [5,]   -0.1099249 0.03624808
```

```r
CIpercentile <- cbind(apply(betaboot, 2, quantile, probs = alpha/2),
    apply(betaboot, 2, quantile, probs = 1 - alpha/2))
print(CIpercentile)
```

```
##              [,1]       [,2]
## [1,] -221.4820617 5.61483854
## [2,]    0.2249939 6.28848895
## [3,]    0.1545856 0.33876000
## [4,]   -0.2986055 0.16237066
## [5,]   -0.1089141 0.03646585
```

```r
beta_full <- coef(quant_model)
CIpivotal <- cbind(2 * beta_full - apply(betaboot, 2, quantile, probs = 1 -
    alpha/2), 2 * beta_full - apply(betaboot, 2, quantile, probs = alpha/2))
print(CIpivotal)
```

```
##                           [,1]        [,2]
## (Intercept)       -231.5750365 -4.47813629
## temperature          0.5097786  6.57327363
## pulse                0.1662342  0.35040860
## packed_cell_volume  -0.3010173  0.15995882
```

```
## total_protein        -0.1107758  0.03460421
```

```r
summary(quant_model)$coefficients[, "Std. Error"]
```

```
##       (Intercept)       temperature             pulse packed_cell_volume
##       52.66872917        1.39221025        0.04231417         0.10832383
##     total_protein
##        0.03675915
```

The coefficient estimates are essentially the same across methods (OLS / jackknife mean / bootstrap mean). Both resampling methods give very different estimates than OLS. The Jackknife SEs and bootstrap SEs are very similar. These SEs are much larger than OLS SEs. The two sets of intervals are very similar to one another, further supporting the stability of the bootstrap results. Compared to OLS confidence intervals, the bootstrap intervals are significantly wider.

□

Problem 3. *Proof.*

```r
# Nonparametric test of staistical significant: Permutation F Test

f = summary(quant_model)$fstatistic[1]
print(f)
```

```
##    value
## 13.92634
```

```r
pvalue = pf(f, p, n - p - 1, lower.tail = FALSE)
print(pvalue)
```

```
##        value
## 6.431426e-12
```

```r
nperm = 10000
fperm = numeric(nperm)
for (k in 1:nperm) {
    Yperm = sample(Y)
    perm = lm(respiratory_rate ~ temperature + pulse + packed_cell_volume +
        total_protein, horse)
    fperm[k] = summary(perm)$fstatistic[1]
}
pvalue = (1 + sum(fperm > f))/(1 + nperm)
print(pvalue)
```

```
## [1] 9.999e-05
```

The estimated F test p-value through the permutation method is equal to 1e-4. Since the pvalue is less than alpha, we can reject the null hypothesis. The pvalue under the theoretical null distribution is 6.4e-12, which is smaller than the pvalue from the permutation test. However, they both lead to the rejection of the null.

```
# Nonparametric test of statistical significance for the effect of
# temperature Permutation t Test

t = summary(quant_model)$coefficients["temperature", 3]
print(t)
```

```
## [1] 2.441538
```

```
pvalue = 2 * pt(abs(t), n - p - 1, lower.tail = FALSE)
print(pvalue)
```

```
## [1] 0.01536693
```

```
aux = lm(respiratory_rate ~ pulse + packed_cell_volume + total_protein,
    horse)
fitted = aux$fitted.values
residuals = aux$residuals
tperm = numeric(nperm)
for (k in 1:nperm) {
    Yperm = fitted + sample(residuals)
    perm = lm(Yperm ~ temperature + pulse + packed_cell_volume + total_protein,
        horse)
    tperm[k] = summary(perm)$coefficients["temperature", 3]
}
pvalue = (1 + sum(abs(tperm) > abs(t)))/(1 + nperm)
print(pvalue)
```

```
## [1] 0.01669833
```

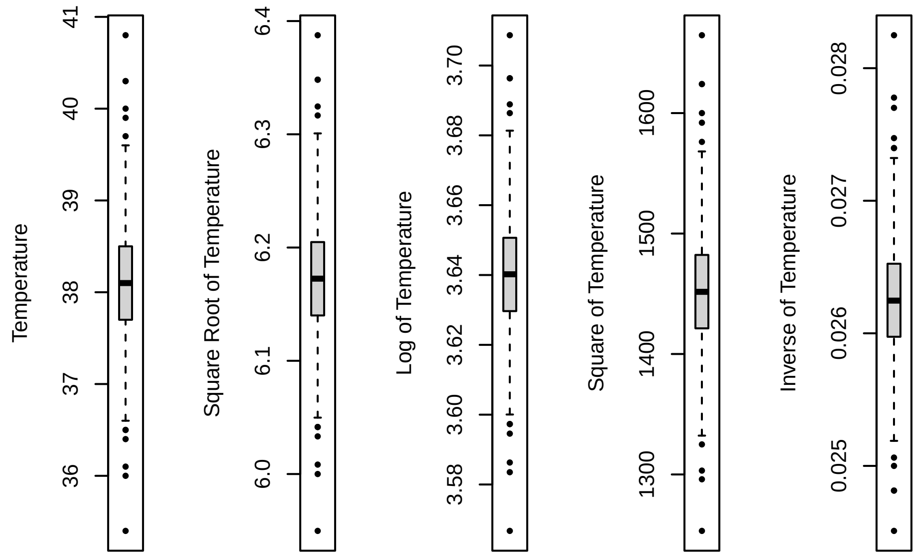
The estimated t test p-value through the permutation method is equal to 0.017. Since the pvalue is less than alpha, we can reject the null hypothesis. The pvalue under the theoretical null distribution is 0.015, which is very similar to the pvaqlue obtained from the permutation test.

□

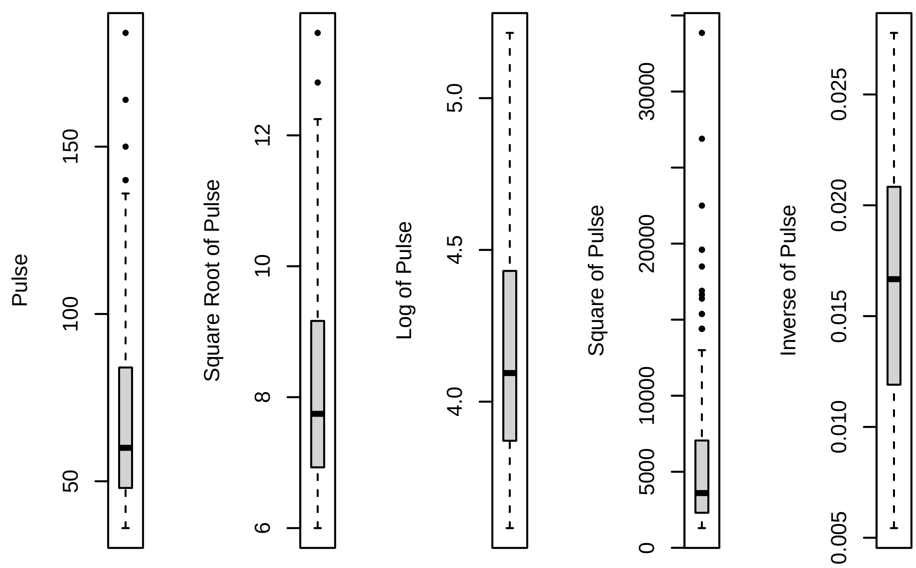
Problem 4. *Proof.*

```
# Check for possible covariate transformations

par(mfrow = c(1, 5))
boxplot(horse$temperature, ylab = "Temperature", pch = 16)
boxplot(sqrt(horse$temperature), ylab = "Square Root of Temperature",
    pch = 16)
boxplot(log(horse$temperature), ylab = "Log of Temperature", pch = 16)
boxplot((horse$temperature)^2, ylab = "Square of Temperature", pch = 16)
```

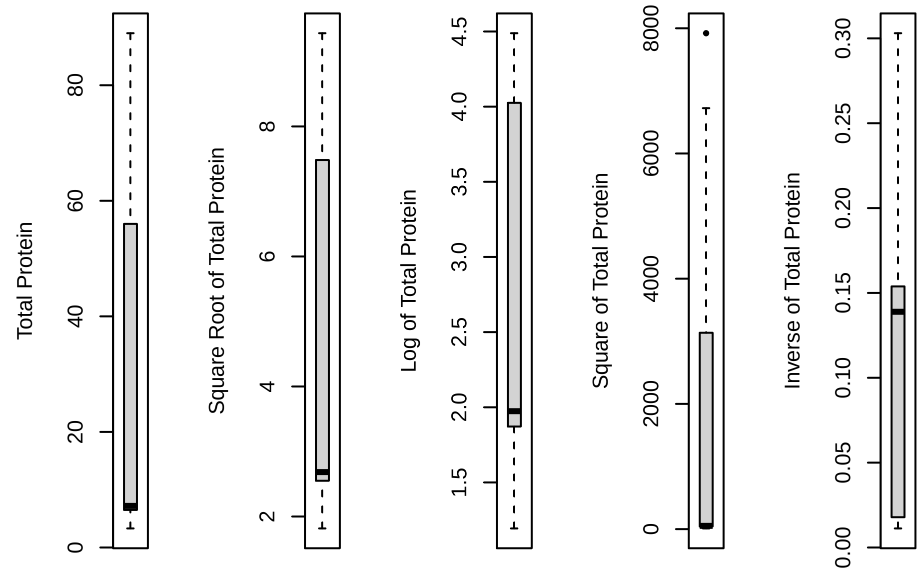
```r
boxplot((horse$temperature)^(-1), ylab = "Inverse of Temperature", pch = 16)
```



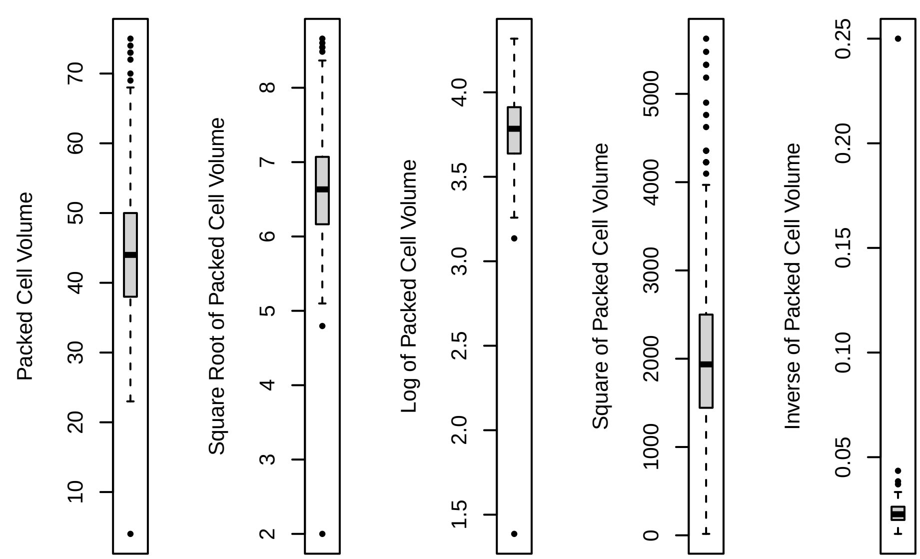

```r
boxplot(horse$pulse, ylab = "Pulse", pch = 16)
boxplot(sqrt(horse$pulse), ylab = "Square Root of Pulse", pch = 16)
boxplot(log(horse$pulse), ylab = "Log of Pulse", pch = 16)
boxplot((horse$pulse)^2, ylab = "Square of Pulse", pch = 16)
boxplot((horse$pulse)^(-1), ylab = "Inverse of Pulse", pch = 16)
```




```r
boxplot(horse$total_protein, ylab = "Total Protein", pch = 16)
boxplot(sqrt(horse$total_protein), ylab = "Square Root of Total Protein",
    pch = 16)
boxplot(log(horse$total_protein), ylab = "Log of Total Protein", pch = 16)
boxplot((horse$total_protein)^2, ylab = "Square of Total Protein", pch = 16)
boxplot((horse$total_protein)^(-1), ylab = "Inverse of Total Protein",
    pch = 16)
```

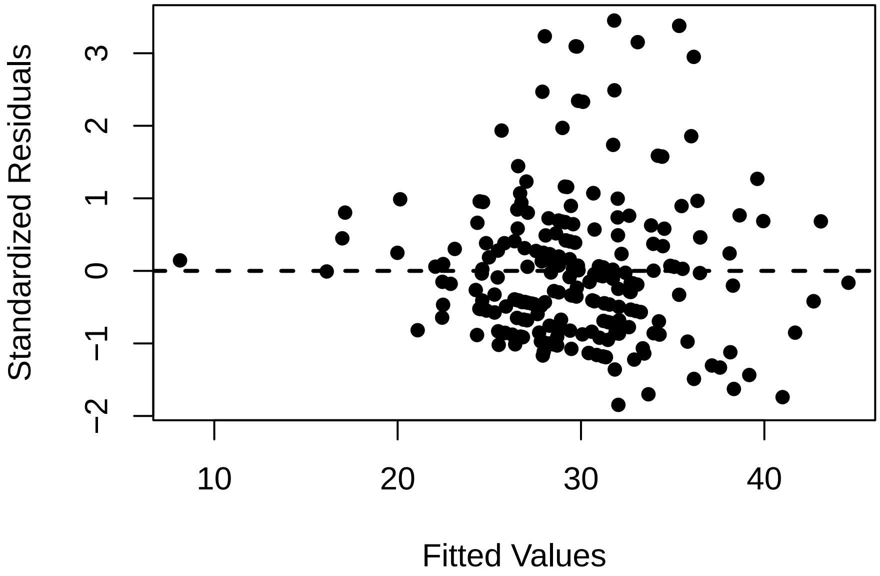



```r
boxplot(horse$packed_cell_volume, ylab = "Packed Cell Volume", pch = 16)
boxplot(sqrt(horse$packed_cell_volume), ylab = "Square Root of Packed Cell Volume",
    pch = 16)
boxplot(log(horse$packed_cell_volume), ylab = "Log of Packed Cell Volume",
    pch = 16)
boxplot((horse$packed_cell_volume)^2, ylab = "Square of Packed Cell Volume",
    pch = 16)
boxplot((horse$packed_cell_volume)^(-1), ylab = "Inverse of Packed Cell Volume",
    pch = 16)
```



Pulse and total protein seem skewed, so need transformations. For pulse, the best transformation is inverse. For total protein, the best transformation is a log transformation. A log transformation is reasonable for packed cell volume to decrease outliers and make for a more symmetric distribution.

```r
fit = lm(respiratory_rate ~ temperature + (1/pulse) + log(packed_cell_volume) +
    log(total_protein), horse)
plot(fit$fitted.values, rstandard(fit), xlab = "Fitted Values", ylab = "Standardized Res
    pch = 16)
abline(h = 0, lty = 2, lwd = 2)
```



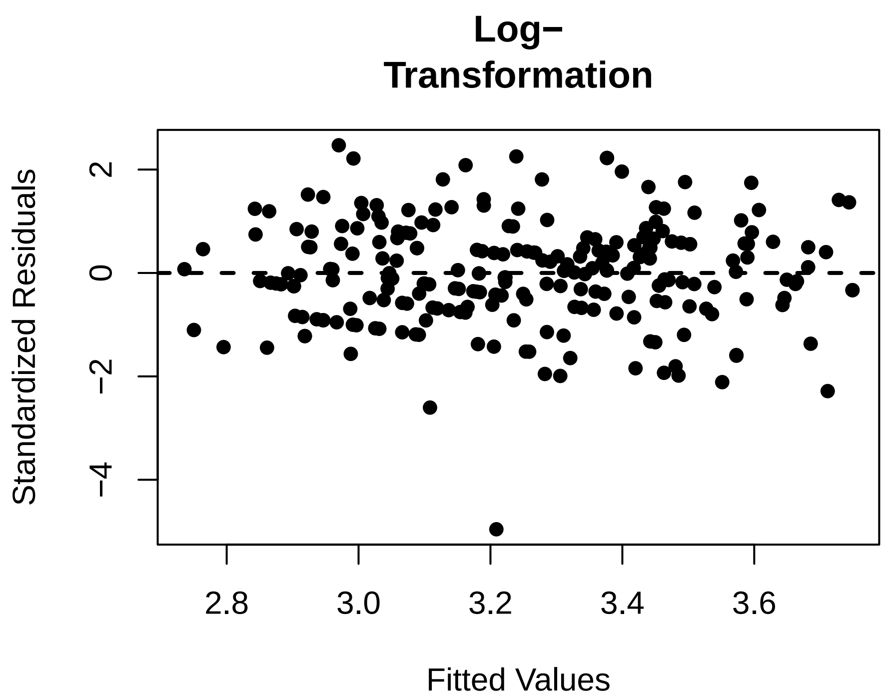
Since constant variance is violated, a transformation should be applied to the response variable.

```r
# Box-Cox transformation
horse$inv_pulse = 1/horse$pulse
Y = horse$respiratory_rate

lambda = seq(-2, 2, 0.01)
loglik = numeric(401)
for (i in 1:401) {
    if (lambda[i] == 0) {
        Ypower = log(Y)
    } else {
        Ypower = (Y^lambda[i] - 1)/lambda[i]
    }
    power = lm(Ypower ~ temperature + inv_pulse + log(packed_cell_volume) +
        log(total_protein), horse)
    loglik[i] = logLik(power)[1] + (lambda[i] - 1) * sum(log(Y))
}

CI = range(lambda[loglik > max(loglik) - qchisq(alpha, 1, lower.tail = FALSE)/2])
print(CI)
```

```
## [1] 0.02 0.34
```

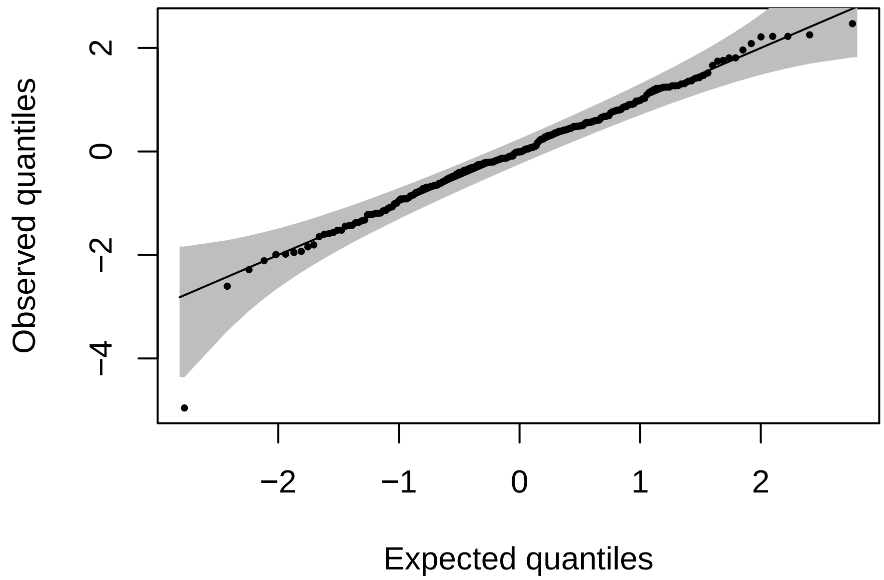
Since the CI is close to covering 0, a log transformation is reasonable.

```
power = lm(log(respiratory_rate) ~ temperature + inv_pulse + log(packed_cell_volume) +
    log(total_protein), horse)
plot(power$fitted.values, rstandard(power), main = "Log-
Transformation",
    xlab = "Fitted Values", ylab = "
Standardized Residuals", pch = 16)
abline(h = 0, lty = 2, lwd = 2)
```




```
library(qqconf)
qq_conf_plot(rstandard(power), points_params = list(pch = 16, cex = 0.5))
```

```
## no dparams supplied. Estimating parameters from the data...
```



All standardized residuals fall within the band, except one. This is enough to reject normality.

□

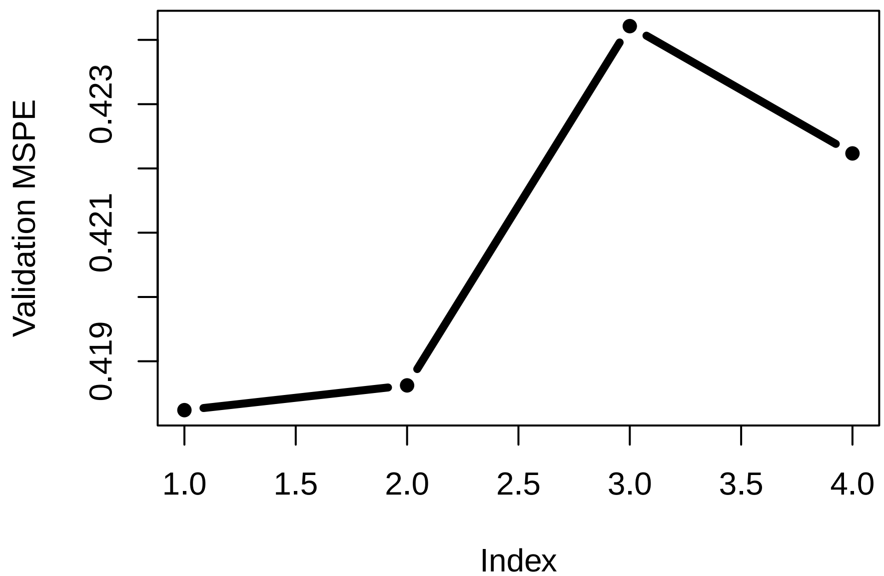
Problem 5. *Proof.*

```r
# Best subset selection with valid post-selection inference

library(leaps)

# New horse data frame with 5 transformed variables
horse_transformed = data.frame(Y = log(horse$respiratory_rate), temperature = horse$temp
    inv_pulse = 1/horse$pulse, log_pcv = log(horse$packed_cell_volume),
    log_tp = log(horse$total_protein))

Y = horse_transformed$Y
X = horse_transformed[, -1]
n = nrow(X)
P = ncol(X)

train = sample(n, 0.6 * n)
valid = sample(setdiff(1:n, train), 0.2 * n)
test = setdiff(1:n, c(train, valid))
MSPE = numeric(P)

full = lm(Y ~ ., horse_transformed, train)
sub = regsubsets(formula(full), horse_transformed[train, ], nvmax = P)
summary(sub)$which
```

```
##   (Intercept) temperature inv_pulse log_pcv log_tp
## 1        TRUE       FALSE      TRUE   FALSE  FALSE
```

```
## 2         TRUE         TRUE       TRUE    FALSE   FALSE
## 3         TRUE         TRUE       TRUE     TRUE   FALSE
## 4         TRUE         TRUE       TRUE     TRUE    TRUE
```

```r
for (p in 1:P) {
    fit = lm(Y ~ ., X[, summary(sub)$which[p, -1], drop = FALSE], train)
    preds = predict(fit, X[valid, summary(sub)$which[p, -1], drop = FALSE])
    MSPE[p] = mean((Y[valid] - preds)^2)
}

plot(MSPE, type = "b", pch = 16, lwd = 4, ylab = "Validation MSPE")
```




```r
p = which.min(MSPE)
final_model = lm(Y ~ ., X[, summary(sub)$which[p, -1], drop = FALSE],
    test)
summary(final_model)
```

```
##
## Call:
## lm(formula = Y ~ ., data = X[, summary(sub)$which[p, -1], drop = FALSE],
##     subset = test)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.92884 -0.24592  0.00328  0.24767  1.19785
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   3.8510     0.2043  18.846   <2e-16 ***
## inv_pulse   -40.0116    12.5067  -3.199   0.0025 **
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4745 on 46 degrees of freedom
## Multiple R-squared:  0.182,  Adjusted R-squared:  0.1642
## F-statistic: 10.23 on 1 and 46 DF,  p-value: 0.002497
```

Based on the validation MSPE curve, the full (4 covariates) model provides the most accurate predictor for respiratory rate. Among all of the predictors, inverse pulse (inv_pulse) is the only statistically significant variable on the test set and shows that horses with higher pulse rates tend to have higher respiratory rates. The remaining predictors (temperature, log(packed cell volume), and log(total protein)) are not individually statistically significant.

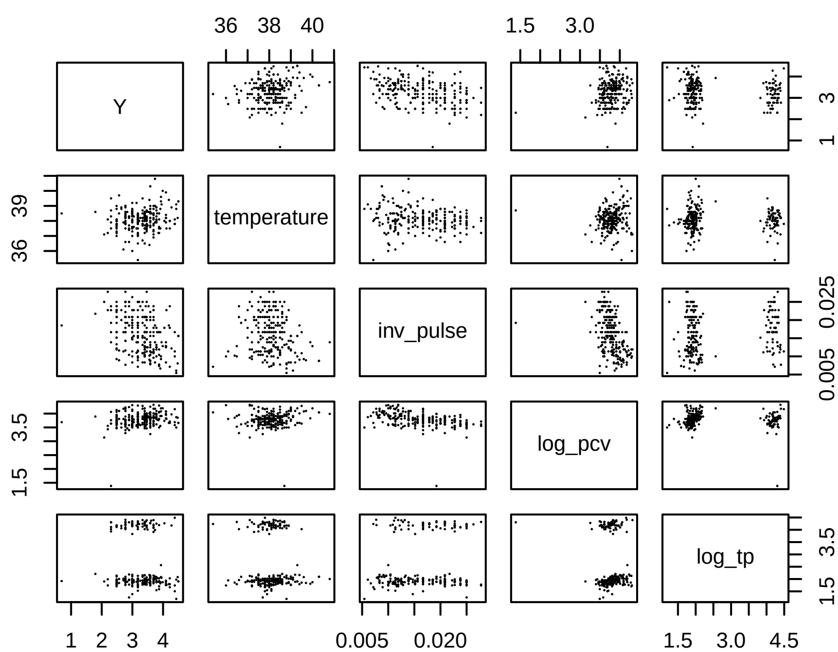
□

Problem 6. *Proof.*

```r
# Investigate collinearity
library(faraway)
cor(horse_transformed)
```

```
##                      Y temperature  inv_pulse    log_pcv      log_tp
## Y            1.0000000  0.16920982 -0.4060674  0.2176732 -0.11911463
## temperature  0.1692098  1.00000000 -0.1751922  0.1032828 -0.04926768
## inv_pulse   -0.4060674 -0.17519222  1.0000000 -0.4159228  0.14147543
## log_pcv      0.2176732  0.10328283 -0.4159228  1.0000000 -0.09311020
## log_tp      -0.1191146 -0.04926768  0.1414754 -0.0931102  1.00000000
```

```r
plot(horse_transformed, pch = 16, cex = 0.25)
```



```r
cat("VIF:", vif(full))
```

```
## VIF: 1.019564 1.249903 1.235599 1.061112
```

Based on the VIFs and correlation plots, the transformed quantitative variables aren't very collinear.

```r
# Ridge Regression
library(glmnet)
train_boot = c(train, valid)
X = as.matrix(X)
cv.ridge = cv.glmnet(X[train_boot, ], Y[train_boot], nfolds = 6, alpha = 0)
ridge = glmnet(X[test, ], Y[test], alpha = 0, lambda = cv.ridge$lambda.min)
print(ridge$beta)
```

```
## 4 x 1 sparse Matrix of class "dgCMatrix"
##                         s0
## temperature   0.065253640
## inv_pulse   -26.193003155
## log_pcv       0.135120805
## log_tp       -0.008200548
```

```r
# Bootstrap
nboot = 10000
beta = matrix(0, nboot, P)
colnames(beta) = colnames(X)
for (i in 1:nboot) {
    ind = sample(test, replace = TRUE)
    beta[i, ] = glmnet(X[ind, ], Y[ind], alpha = 0, lambda = cv.ridge$lambda.min)$beta[,
        ]
}
SE = apply(beta, 2, sd)
cat("SE after Bootstrap:", SE, "\n")
```

```
## SE after Bootstrap: 0.06478434 9.057431 0.2336487 0.05209877
```

```r
ols = lm(Y ~ ., horse_transformed[train, ])
cat("Summary of OLS coefficients for comparison:", "\n")
```

```
## Summary of OLS coefficients for comparison:
```

```r
summary(ols)$coefficients
```

```
##                  Estimate Std. Error    t value     Pr(>|t|)
## (Intercept)   -0.21799383 2.11256553 -0.1031891 9.179616e-01
## temperature    0.09222970 0.05287914  1.7441603 8.334173e-02
## inv_pulse    -35.71734705 8.46762134 -4.2181087 4.417755e-05
## log_pcv        0.17708872 0.14653704  1.2084912 2.289102e-01
```

```
## log_tp       -0.04042411 0.03929397 -1.0287612 3.053787e-01
```

The ridge regression coefficients all shrunk closer to zero in comparison to the OLS coefficients. The bootstrap results SEs are slightly higher than the SEs calculated using OLS.

□

Problem 7. *Proof.*

```r
# Lasso Regression
cv.lasso = cv.glmnet(X[train, ], Y[train], alpha = 1)
lasso = glmnet(X[test, ], Y[test], nfolds = 6, alpha = 1, lambda = cv.lasso$lambda.min)
print(lasso$beta)
```

```
## 4 x 1 sparse Matrix of class "dgCMatrix"
##                        s0
## temperature   0.06868708
## inv_pulse   -34.98917710
## log_pcv       0.02454834
## log_tp         .
```

```r
for (i in 1:nboot) {
    ind = sample(test, replace = TRUE)
    beta[i, ] = glmnet(X[ind, ], Y[ind], alpha = 1, lambda = cv.lasso$lambda.min)$beta[,
        ]
}
SE = apply(beta, 2, sd)
cat("SE after Bootstrap:", SE, "\n")
```

```
## SE after Bootstrap: 0.08517503 13.86536 0.2958224 0.06885285
```

While some of the estimated lasso coefficients become extremely close to zero, others stay the same. Because the quantitative variables aren't collinear, the estimated SEs of the lasso coefficients are greater than the SEs estimated by the OLS. □

Problem 8. *Proof.*

```r
# Investigate the interaction effects between pain and all
# available quantitative covariates on the response variable.

# Factor qualitative variable
horse$pain = as.factor(horse$pain)

Y = log(horse$respiratory_rate)
inv_pulse = 1/horse$pulse
log_pcv = log(horse$packed_cell_volume)
log_tp = log(horse$total_protein)
horse$pain = as.factor(horse$pain)
```

```r
fit_interact = lm(Y ~ pain * (temperature + inv_pulse + log_pcv + log_tp),
    horse)
anova(fit_interact)
```

```
## Analysis of Variance Table
##
## Response: Y
##                   Df Sum Sq Mean Sq F value    Pr(>F)
## pain               4  5.449  1.3622  5.0559 0.0006506 ***
## temperature        1  1.381  1.3812  5.1262 0.0245650 *
## inv_pulse          1  6.688  6.6882 24.8230  1.29e-06 ***
## log_pcv            1  0.113  0.1131  0.4198 0.5177107
## log_tp             1  0.278  0.2776  1.0305 0.3111925
## pain:temperature   4  1.199  0.2997  1.1122 0.3516623
## pain:inv_pulse     4  0.547  0.1367  0.5075 0.7302813
## pain:log_pcv       4  0.620  0.1549  0.5750 0.6810488
## pain:log_tp        4  0.310  0.0776  0.2880 0.8855690
## Residuals        215 57.928  0.2694
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
fit_interact = update(fit_interact, . ~ . - pain:log_tp)
anova(fit_interact)
```

```
## Analysis of Variance Table
##
## Response: Y
##                   Df Sum Sq Mean Sq F value    Pr(>F)
## pain               4  5.449  1.3622  5.1226 0.0005785 ***
## temperature        1  1.381  1.3812  5.1937 0.0236322 *
## inv_pulse          1  6.688  6.6882 25.1501 1.095e-06 ***
## log_pcv            1  0.113  0.1131  0.4254 0.5149551
## log_tp             1  0.278  0.2776  1.0440 0.3080126
## pain:temperature   4  1.199  0.2997  1.1269 0.3446763
## pain:inv_pulse     4  0.547  0.1367  0.5142 0.7253855
## pain:log_pcv       4  0.620  0.1549  0.5826 0.6755637
## Residuals        219 58.239  0.2659
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
fit_interact = update(fit_interact, . ~ . - pain:inv_pulse)
anova(fit_interact)
```

```
## Analysis of Variance Table
##
```

```
## Response: Y
##                   Df Sum Sq Mean Sq F value    Pr(>F)
## pain               4  5.449  1.3622  5.1754 0.0005264 ***
## temperature        1  1.381  1.3812  5.2473 0.0229151 *
## inv_pulse          1  6.688  6.6882 25.4095 9.584e-07 ***
## log_pcv            1  0.113  0.1131  0.4298 0.5127852
## log_tp             1  0.278  0.2776  1.0548 0.3055163
## pain:temperature   4  1.199  0.2997  1.1385 0.3392033
## pain:log_pcv       4  0.708  0.1771  0.6728 0.6114676
## Residuals        223 58.697  0.2632
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
fit_interact = update(fit_interact, . ~ . - pain:log_pcv)
anova(fit_interact)
```

```
## Analysis of Variance Table
##
## Response: Y
##                   Df Sum Sq Mean Sq F value    Pr(>F)
## pain               4  5.449  1.3622  5.2054 0.0004975 ***
## temperature        1  1.381  1.3812  5.2777 0.0225109 *
## inv_pulse          1  6.688  6.6882 25.5569 8.842e-07 ***
## log_pcv            1  0.113  0.1131  0.4322 0.5115548
## log_tp             1  0.278  0.2776  1.0609 0.3041019
## pain:temperature   4  1.199  0.2997  1.1451 0.3361034
## Residuals        227 59.405  0.2617
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
fit_interact = update(fit_interact, . ~ . - log_pcv)
anova(fit_interact)
```

```
## Analysis of Variance Table
##
## Response: Y
##                   Df Sum Sq Mean Sq F value    Pr(>F)
## pain               4  5.449  1.3622  5.2263 0.0004797 ***
## temperature        1  1.381  1.3812  5.2989 0.0222414 *
## inv_pulse          1  6.688  6.6882 25.6593 8.405e-07 ***
## log_tp             1  0.296  0.2957  1.1345 0.2879454
## pain:temperature   4  1.270  0.3176  1.2184 0.3037703
## Residuals        228 59.429  0.2607
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
fit_interact = update(fit_interact, . ~ . - pain:temperature)
anova(fit_interact)
```

```
## Analysis of Variance Table
##
## Response: Y
##              Df Sum Sq Mean Sq F value    Pr(>F)
## pain          4  5.449  1.3622  5.2067 0.0004928 ***
## temperature   1  1.381  1.3812  5.2790 0.0224749 *
## inv_pulse     1  6.688  6.6882 25.5631  8.69e-07 ***
## log_tp        1  0.296  0.2957  1.1302 0.2888297
## Residuals   232 60.699  0.2616
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
fit_interact = update(fit_interact, . ~ . - log_tp)
anova(fit_interact)
```

```
## Analysis of Variance Table
##
## Response: Y
##              Df Sum Sq Mean Sq F value    Pr(>F)
## pain          4  5.449  1.3622  5.2038 0.0004944 ***
## temperature   1  1.381  1.3812  5.2761 0.0225083 *
## inv_pulse     1  6.688  6.6882 25.5488 8.723e-07 ***
## Residuals   233 60.995  0.2618
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
summary(fit_interact)
```

```
##
## Call:
## lm(formula = Y ~ pain + temperature + inv_pulse, data = horse)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.49769 -0.28990  0.01148  0.30445  1.25982
##
## Coefficients:
##                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)      0.91073    1.87891   0.485    0.628
## paincontinuous   0.17101    0.13523   1.265    0.207
## paindepressed    0.12539    0.11199   1.120    0.264
## painmild         0.07588    0.09939   0.763    0.446
## painsevere       0.09022    0.12632   0.714    0.476
```

```
## temperature      0.07486    0.04863   1.539    0.125
## inv_pulse      -36.61249    7.24343  -5.055 8.72e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.5116 on 233 degrees of freedom
## Multiple R-squared:  0.1814, Adjusted R-squared:  0.1603
## F-statistic: 8.607 on 6 and 233 DF,  p-value: 1.84e-08
```

Final Model:

$$Y = \beta_0+\beta_1\,\text{pain}_{\text{continuous}}+\beta_2\,\text{pain}_{\text{depressed}}+\beta_3\,\text{pain}_{\text{mild}}+\beta_4\,\text{pain}_{\text{severe}}+\beta_5\,\text{temperature}+\beta_6\,\text{inv\_pulse}+\varepsilon$$

Interpretation of each covariate in the final model:

pain (continuous): Horses in the continuous pain category have an expected log-respiratory-rate that is 0.171 units higher than horses in the reference pain category, holding temperature and inverse pulse constant.

pain (depressed): Horses rated as depressed have an expected log-respiratory-rate that is 0.126 units higher than horses in the reference pain category, adjusting for temperature and inverse pulse.

pain (mild): Horses with mild pain have an expected log-respiratory-rate that is 0.076 units higher than the reference category, controlling for temperature and inverse pulse.

pain (severe): Horses with severe pain have an expected log-respiratory-rate that is 0.090 units higher than horses in the reference category, holding temperature and inverse pulse constant.

temperature: For each 1-degree increase in temperature, the expected log-respiratory-rate increases by 0.0749 units, controlling for pain level and inverse pulse.

inverse pulse (inv_pulse): For each 1-unit increase in inverse pulse (which corresponds to a decrease in pulse rate), the expected log-respiratory-rate decreases by 36.61 units, holding pain level and temperature constant.

□

Problem 9. *Proof.*

```r
# Mixed-effects Model

Y = horse$respiratory_rate
horse$pain = factor(horse$pain)
horse$surgical_lesion = factor(horse$surgical_lesion)
horse$peristalsis = factor(horse$peristalsis)
horse$abdominal_distension = factor(horse$abdominal_distension)
horse$peripheral_pulse = factor(horse$peripheral_pulse)
horse$temperature_of_extremities = factor(horse$temperature_of_extremities)
```

```r
horse$surgery <- factor(horse$surgery)
horse$outcome <- factor(horse$outcome)
horse$age <- factor(horse$age)
```

```r
# chosen random effects

# Used transformed variables from problem 8 for reference

library(lme4)
mixed_eff_mod = lmer(Y ~ temperature + inv_pulse + log_pcv + log_tp +
    outcome + surgery + abdominal_distension + pain + (1 | temperature_of_extremities) +
    (1 | peripheral_pulse) + (1 | age) + (1 | surgical_lesion) + (1 |
    peristalsis), data = horse)
```

```r
sigma = data.frame(summary(mixed_eff_mod)$varcor)$sdcor
names(sigma) = data.frame(summary(mixed_eff_mod)$varcor)$grp
```

```r
ICC = sigma^2/sum(sigma^2)
print(ICC)
```

```
##                peristalsis           peripheral_pulse
##                0.003159587                0.159798515
## temperature_of_extremities            surgical_lesion
##                0.014265004                0.014474495
##                        age                   Residual
##                0.435440517                0.372861882
```

```r
summary(mixed_eff_mod, correlation = FALSE)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: Y ~ temperature + inv_pulse + log_pcv + log_tp + outcome + surgery +
##     abdominal_distension + pain + (1 | temperature_of_extremities) +
##     (1 | peripheral_pulse) + (1 | age) + (1 | surgical_lesion) +
##     (1 | peristalsis)
##    Data: horse
##
## REML criterion at convergence: 1893.4
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -1.9999 -0.6687 -0.1350  0.5532  3.1234
##
## Random effects:
##  Groups                     Name        Variance Std.Dev.
##  peristalsis                (Intercept)  1.706   1.306
##  peripheral_pulse           (Intercept) 86.274   9.288
```

```
##  temperature_of_extremities (Intercept)   7.702   2.775
##  surgical_lesion            (Intercept)   7.815   2.795
##  age                        (Intercept) 235.092  15.333
##  Residual                              201.306  14.188
## Number of obs: 240, groups:
## peristalsis, 4; peripheral_pulse, 4; temperature_of_extremities, 4; surgical_lesion,
##
## Fixed effects:
##                              Estimate Std. Error t value
## (Intercept)                  -78.5493    59.6465  -1.317
## temperature                    2.7916     1.4465   1.930
## inv_pulse                   -255.3853   265.0700  -0.963
## log_pcv                        3.3211     4.2159   0.788
## log_tp                        -1.3678     0.9921  -1.379
## outcomeeuthanized              3.3997     3.9336   0.864
## outcomelived                   3.5648     2.9265   1.218
## surgery1                       2.8395     2.4341   1.167
## abdominal_distensionnone       1.6184     2.7801   0.582
## abdominal_distensionsevere     5.1821     3.6876   1.405
## abdominal_distensionslight     2.3686     2.8540   0.830
## paincontinuous                 4.3930     4.3918   1.000
## paindepressed                  7.5573     3.4205   2.209
## painmild                       2.7623     3.0152   0.916
## painsevere                     1.3234     3.9828   0.332
```

```r
anova(mixed_eff_mod)
```

```
## Analysis of Variance Table
##                        npar  Sum Sq Mean Sq F value
## temperature               1 2123.98 2123.98 10.5510
## inv_pulse                 1  674.08  674.08  3.3485
## log_pcv                   1  188.31  188.31  0.9355
## log_tp                    1  133.83  133.83  0.6648
## outcome                   2  482.84  241.42  1.1993
## surgery                   1  270.52  270.52  1.3438
## abdominal_distension      3  546.77  182.26  0.9054
## pain                      4 1271.03  317.76  1.5785
```

```r
# Box Cox
Y = horse$respiratory_rate

lambda = seq(-2, 2, 0.01)
loglik = numeric(401)
for (i in 1:401) {
    if (lambda[i] == 0) {
        Ypower = log(Y)
```

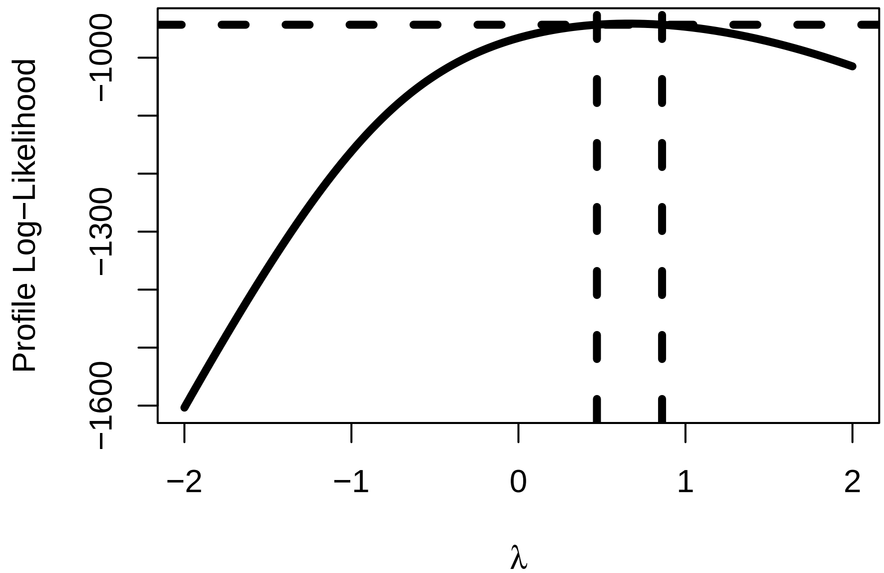
```r
    } else {
        Ypower = (Y^lambda[i] - 1)/lambda[i]
    }
    power = lmer(Ypower ~ temperature + inv_pulse + log_pcv + log_tp +
        outcome + surgery + abdominal_distension + pain + (1 | temperature_of_extremitie
        (1 | peripheral_pulse) + (1 | age) + (1 | surgical_lesion) +
        (1 | peristalsis), horse)
    loglik[i] = logLik(power)[1] + (lambda[i] - 1) * sum(log(Y))
}
```

```
## Warning in checkConv(attr(opt, "derivs"), opt$par, ctrl = control$checkConv, :
## unable to evaluate scaled gradient

## Warning in checkConv(attr(opt, "derivs"), opt$par, ctrl = control$checkConv, :
## Model failed to converge: degenerate Hessian with 1 negative eigenvalues

## Warning in checkConv(attr(opt, "derivs"), opt$par, ctrl = control$checkConv, :
## unable to evaluate scaled gradient

## Warning in checkConv(attr(opt, "derivs"), opt$par, ctrl = control$checkConv, :
## Model failed to converge: degenerate Hessian with 1 negative eigenvalues
```

```r
alpha = 0.05
CI = range(lambda[loglik > max(loglik) - qchisq(alpha, 1, lower.tail = FALSE)/2])
print(CI)
```

```
## [1] 0.47 0.86
```

```r
plot(lambda, loglik, "l", xlab = expression(lambda), ylab = "
Profile Log-Likelihood",
    lwd = 4)
abline(h = max(loglik) - qchisq(0.95, 1)/2, lty = 2, lwd = 4)
abline(v = CI, lty = 2, lwd = 4)
```



The covariates chosen as random effects are age, peripheral_pulse, peristalsis, and temperature_of_extremities, since these were the categorical variables with the largest nonzero variance components in the mixed-effects model. All categorical variables with exactly zero variance components were treated as fixed effects, as instructed.

A suitable transformation for our response variable is.

The total variation in your response variable explained by each random effect is as follows: abdominal_distension (variance = 0.02668, 2.67%) and temperature_of_extremities (variance = 0.00652, 0.65%), and surgery (variance = 0.005579, 0.56%).

Another interpretation for each ICC: ICC(abdominal_distension) = 0.0936 Two horses with the same abdominal distension level have an expected correlation of about 9.4% in their log respiratory rates. This points too a moderate grouping effect of adominal distension level on respiratory rate.

ICC(temperature_of_extremities) = 0.0229 Two horses with the same extremity temperature category have an expected correlation of about 2.3% in their log respiratory rates. This points to a very small grouping effect of extremity temperature on respiratory rate.

ICC(surgery) = 0.01956768 Two horses that underwent the same type of surgery have an expected correlation of about 2.0% in their log respiratory rates. This points to a very small grouping effect of surgery on respiratory rate.

□

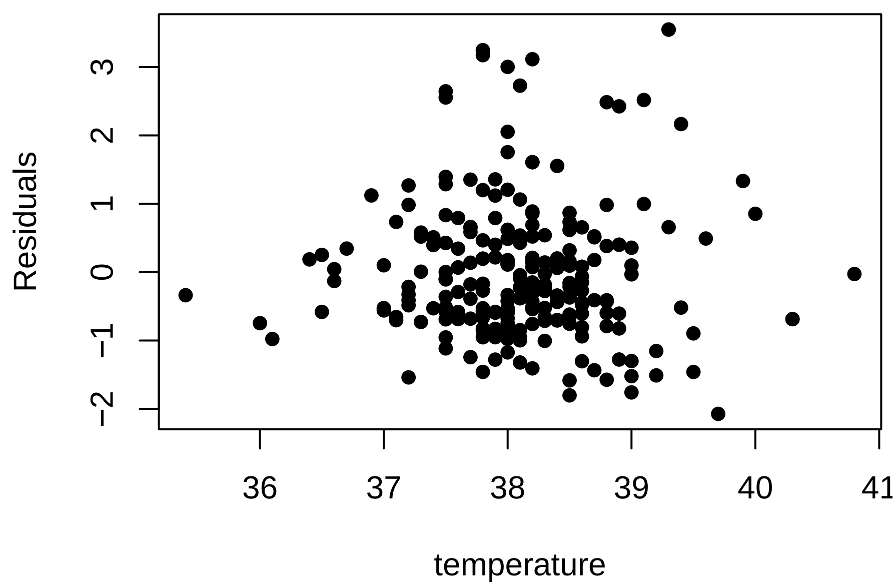
Problem 10. *Proof.*

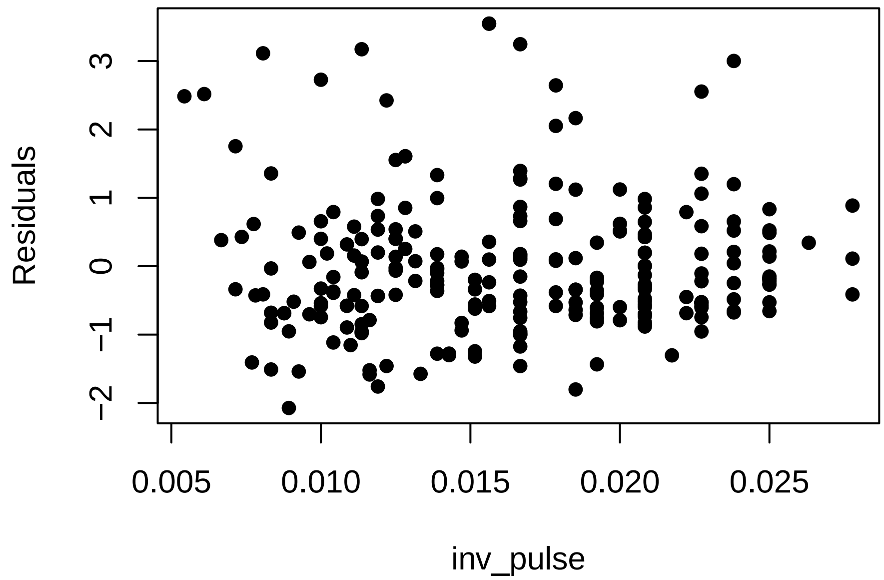
```
base_model <- lm(Y ~ temperature + inv_pulse + log_pcv + log_tp, data = horse)
print(summary(base_model))
```

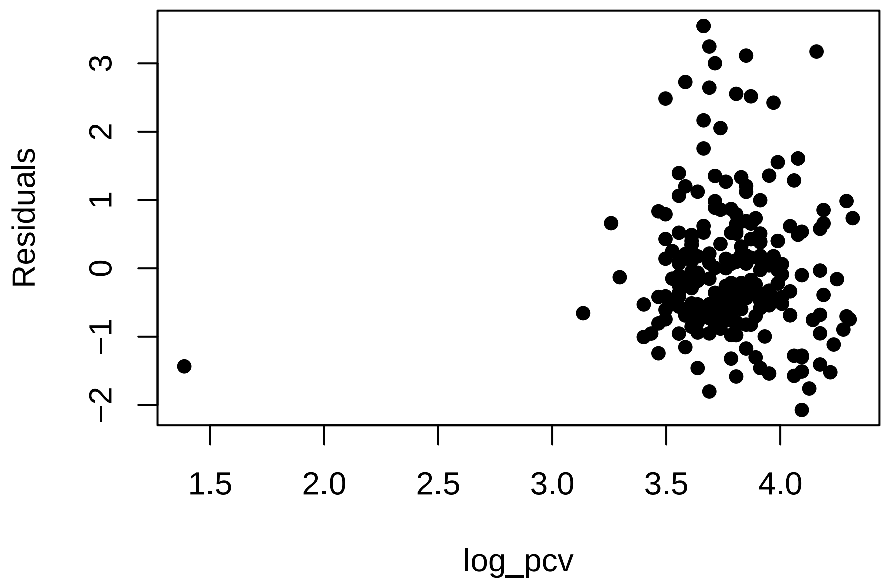
```
##
## Call:
```

```
## lm(formula = Y ~ temperature + inv_pulse + log_pcv + log_tp,
##     data = horse)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -31.539  -9.389  -3.283   6.711  54.318
##
## Coefficients:
##                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -73.7142    56.4478  -1.306   0.1929
## temperature     3.3214     1.4041   2.365   0.0188 *
## inv_pulse   -1126.6554   211.5286  -5.326 2.34e-07 ***
## log_pcv        -0.2385     4.2153  -0.057   0.9549
## log_tp         -1.3970     0.9729  -1.436   0.1523
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 15.45 on 235 degrees of freedom
## Multiple R-squared:  0.1778, Adjusted R-squared:  0.1638
## F-statistic: 12.71 on 4 and 235 DF,  p-value: 2.238e-09
```

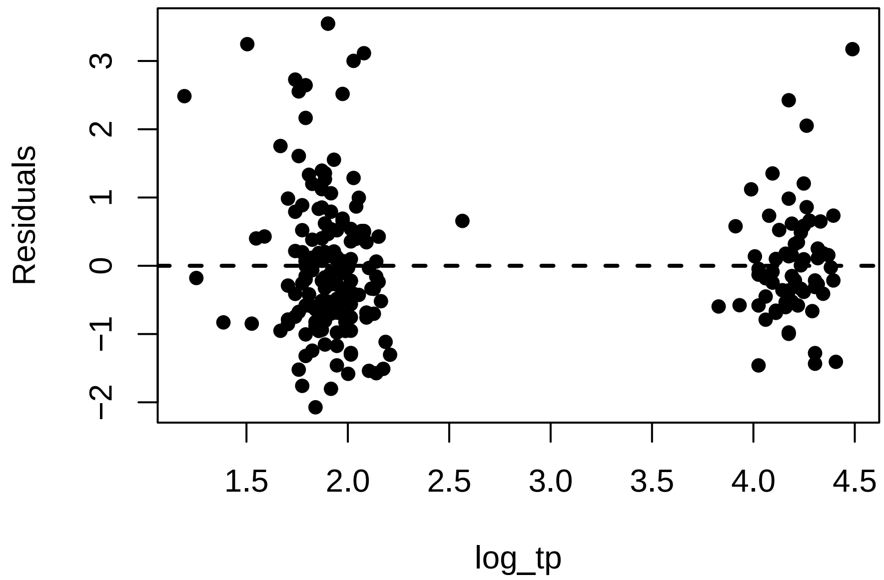
```
Y = log(horse$respiratory_rate)
X = model.matrix(base_model)
n = dim(X)[1]
p = dim(X)[2] - 1
residuals = base_model$residuals
plot(rstandard(base_model) ~ temperature + inv_pulse + log_pcv + log_tp,
    horse, ylab = "Standardized
Residuals", pch = 16)
```








```
abline(h = 0, lty = 2, lwd = 2)
```




```
# Studentized Breusch-Pagan Test
Yaux = residuals(base_model)^2
aux = lm(Yaux ~ temperature + inv_pulse + log_pcv + log_tp, horse)
BP = n * summary(aux)$r.squared
print(BP)
```

```
## [1] 11.83406
```

```
pvalue = pchisq(BP, p, lower.tail = FALSE)
print(pvalue)
```

```
## [1] 0.01862882
```

```
library(lmtest)
bptest(base_model)
```

```
##
##  studentized Breusch-Pagan test
##
## data:  base_model
## BP = 11.834, df = 4, p-value = 0.01863
```

```
# Brown-Forsythe Test -- pain

library(car)
leveneTest(Y ~ pain, data = horse)
```

```
## Levene's Test for Homogeneity of Variance (center = median)
##        Df F value    Pr(>F)
## group   4  5.0709 0.0006161 ***
##       235
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# Brown-Forsythe Test -- outcome
library(car)
leveneTest(Y ~ outcome, data = horse)
```

```
## Levene's Test for Homogeneity of Variance (center = median)
##        Df F value Pr(>F)
## group   2  1.2748 0.2814
##       237
```

```r
# Brown-Forsythe Test -- temp of extremities

library(car)
leveneTest(Y ~ temperature_of_extremities, data = horse)
```

```
## Levene's Test for Homogeneity of Variance (center = median)
##        Df F value Pr(>F)
## group   3  0.9573 0.4136
##       236
```

```r
# Brown-Forsythe Test -- peripheral pulse
library(car)
leveneTest(Y ~ peripheral_pulse, data = horse)
```

```
## Levene's Test for Homogeneity of Variance (center = median)
##        Df F value Pr(>F)
## group   3  2.7338 0.0444 *
##       236
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# Brown-Forsythe Test -- abdominal distension

library(car)
leveneTest(Y ~ abdominal_distension, data = horse)
```

```
## Levene's Test for Homogeneity of Variance (center = median)
##        Df F value Pr(>F)
## group   3  1.7893 0.1499
##       236
```

```r
# Brown-Forsythe Test -- peristalsis

library(car)
leveneTest(Y ~ peristalsis, data = horse)
```

```
## Levene's Test for Homogeneity of Variance (center = median)
##        Df F value  Pr(>F)
```

```
## group   3  2.4855 0.06136 .
##       236
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

For the quantitative variables, a studentized Breusch–Pagan test was conducted. Because the p-value is well above 0.05, we fail to reject the null hypothesis of constant variance. So here is no evidence of heteroscedasticity, and the residual variance does not appear to depend on any of the included quantitative predictors.

For the qualitative varibles, a Brown–Forsythe tests was conducted.

The Brown–Forsythe test yields p = 0.0006161, providing statistically significant evidence that the residual variance differs across pain groups (heteroscedasticity).

The Brown–Forsythe test yields p = 0.2814, so there is no statistically significant evidence of differing variances across outcome groups (homoscedasticity).

The Brown–Forsythe test yields p = 0.4136, so there is no statistically significant evidence that the residual variance differs across peripheral-pulse categories.

The Brown–Forsythe test yields p = 0.0444, indicating statistically significant evidence that residual variances differ across abdominal-distension groups (heteroscedasticity).

The Brown–Forsythe test yields p = 0.061, so there is no statistically significant evidence of variance differences across peristalsis groups at the 5% level. The Brown–Forsythe test yields p = 0.061, so there is no statistically significant evidence that the residual variance differs across peristalsis groups at the 5% level.

□

Problem 11. *Proof.*

```
Y = log(horse$respiratory_rate)
quant_model = lm(Y ~ temperature + inv_pulse + log_pcv + log_tp, horse)

ols_summary = summary(quant_model)

X = model.matrix(quant_model)
n = nrow(X)
p = ncol(X)
beta = coef(quant_model)
residuals = quant_model$residuals
```

```
vcovOLS = summary(quant_model)$sigma^2 * summary(quant_model)$cov.unscaled
vcovOLS
```

```
##              (Intercept)   temperature   inv_pulse       log_pcv        log_tp
## (Intercept)  3.465801376 -8.178581e-02 -3.95918767 -0.0717941594 -4.198632e-03
## temperature -0.081785813  2.144548e-03  0.04609586 -0.0002127801  3.544896e-05
```

```
## inv_pulse   -3.959187671  4.609586e-02 48.66851858  0.3874615003 -2.437183e-02
## log_pcv     -0.071794159 -2.127801e-04  0.38746150  0.0193275056  1.661329e-04
## log_tp      -0.004198632  3.544896e-05 -0.02437183  0.0001661329  1.029491e-03
```

```r
confint(quant_model)
```

```
##                    2.5 %       97.5 %
## (Intercept)  -3.02626049   4.30911449
## temperature  -0.01560159   0.16686712
## inv_pulse   -51.63124263 -24.14315363
## log_pcv      -0.16066009   0.38712271
## log_tp       -0.09471723   0.03170743
```

```r
# Heteroscedasticity-Consistent Estimators


library(sandwich)
vcovHC3 = vcovHC(quant_model, type = "HC3")
vcovHC3
```

```
##              (Intercept)   temperature   inv_pulse       log_pcv        log_tp
## (Intercept)  3.362777937 -7.536590e-02 -3.74147070 -1.139449e-01 -2.075313e-03
## temperature -0.075365899  2.105212e-03  0.01147321 -1.268062e-03  4.728520e-06
## inv_pulse   -3.741470703  1.147321e-02 50.04492660  6.903247e-01 -4.612666e-02
## log_pcv     -0.113944921 -1.268062e-03  0.69032469  3.977325e-02  1.043878e-05
## log_tp      -0.002075313  4.728520e-06 -0.04612666  1.043878e-05  9.672646e-04
```

```r
alpha = 0.05
HC3_CI = cbind(beta - qt(1 - alpha/2, n - p) * sqrt(diag(vcovHC3)), beta +
    qt(1 - alpha/2, n - p) * sqrt(diag(vcovHC3)))

colnames(HC3_CI) = colnames(confint(quant_model))
HC3_CI
```

```
##                    2.5 %       97.5 %
## (Intercept)  -2.97133693   4.25419093
## temperature  -0.01476098   0.16602651
## inv_pulse   -51.82423720 -23.95015906
## log_pcv      -0.27967231   0.50613493
## log_tp       -0.09277705   0.02976725
```

```r
library(lmtest)
confint(coeftest(quant_model, vcov. = vcovHC3))
```

```
##                    2.5 %       97.5 %
## (Intercept)  -2.97133693   4.25419093
## temperature  -0.01476098   0.16602651
## inv_pulse   -51.82423720 -23.95015906
```

```
## log_pcv      -0.27967231   0.50613493
## log_tp       -0.09277705   0.02976725
```

```
test = matrix(0, p, 4)
rownames(test) = names(beta)
colnames(test) = c("Estimate", "Std. Error", "t value", "Pr(>|t|)")

test[, 1] = beta
test[, 2] = sqrt(diag(vcovHC3))
test[, 3] = test[, 1]/test[, 2]
test[, 4] = 2 * pt(abs(test[, 3]), n - p, lower.tail = FALSE)

test
```

```
##                  Estimate Std. Error    t value     Pr(>|t|)
## (Intercept)   0.64142700 1.83378787  0.3497826 7.268152e-01
## temperature   0.07563277 0.04588259  1.6483980 1.006073e-01
## inv_pulse   -37.88719813 7.07424389 -5.3556534 2.028092e-07
## log_pcv       0.11323131 0.19943232  0.5677681 5.707345e-01
## log_tp       -0.03150490 0.03110088 -1.0129908 3.121064e-01
```

```
hc3_test = coeftest(quant_model, vcov. = vcovHC3)
```

```
results_table <- data.frame(Coefficient = rownames(ols_summary$coefficients),
    Estimate = ols_summary$coefficients[, "Estimate"], OLS_SE = ols_summary$coefficients
        "Std. Error"], HC3_SE = sqrt(diag(vcovHC3)), t_HC3 = hc3_test[,
        "t value"], p_HC3 = hc3_test[, "Pr(>|t|)"])

results_table
```

```
##              Coefficient     Estimate     OLS_SE     HC3_SE      t_HC3
## (Intercept) (Intercept)   0.64142700 1.86166629 1.83378787  0.3497826
## temperature temperature   0.07563277 0.04630927 0.04588259  1.6483980
## inv_pulse     inv_pulse -37.88719813 6.97628258 7.07424389 -5.3556534
## log_pcv         log_pcv   0.11323131 0.13902340 0.19943232  0.5677681
## log_tp           log_tp  -0.03150490 0.03208568 0.03110088 -1.0129908
##                     p_HC3
## (Intercept) 7.268152e-01
## temperature 1.006073e-01
## inv_pulse   2.028092e-07
## log_pcv     5.707345e-01
## log_tp      3.121064e-01
```

The HC3 standard errors are larger for most of the coefficients. (Intercept) OLS SE: 1.86166629 HC3 SE: 1.83378787 temperature OLS SE:0.04630927 HC3 SE:0.04588259 pulse OLS SE: 6.97628258 HC3 SE:7.07424389 PCV OLS SE: 0.13902340 HC3 SE: 0.13902340 total_protein OLS SE: 0.03208568 HC3 SE:0.03208568

As the SEs increased, the HC3 t-values got smaller. (Intercept): -2.145 to -1.965 temperature: 2.442 to 2.205 pulse: 5.967 to 5.144 PCV: -0.640 to -0.568 total_protein: -1.011 to -0.972.

The p-values got larger. (Intercept) 0.0330 to 0.0506 (no longer < 0.05) temperature 0.0154 to 0.0284 (still < 0.05) pulse 8.82e-09 to 5.67e-07 (still extremely significant)
PCV 0.5228 to 0.5709 (still NS) total_protein 0.3132 to 0.3319 (still NS)

The intercept is the only coefficient whose significance status changed. In the OLS (Intercept) is significant ($p = 0.033$), but HC3 shows it is only marginally significant ($p = 0.0506$).

The HC3 CIs are consistently wider:

(Intercept) OLS [-216.74, -9.22]; HC3 [-226.25, 0.29]. Which now covers zero. temperature: OLS [0.656,6.142]; HC3 [0.362,6.43] pulse: OLS [0.169,0.336]; HC3 [0.156,0.349] PCV: OLS [-0.283, 0.144]; HC3 [-0.310, 0.171] total_protein: OLS [-0.109, 0.0353]; HC3 [-0.112, 0.038]

The qualitative conclusions remained the same for temperature, pulse, PCV and total_protein.

□

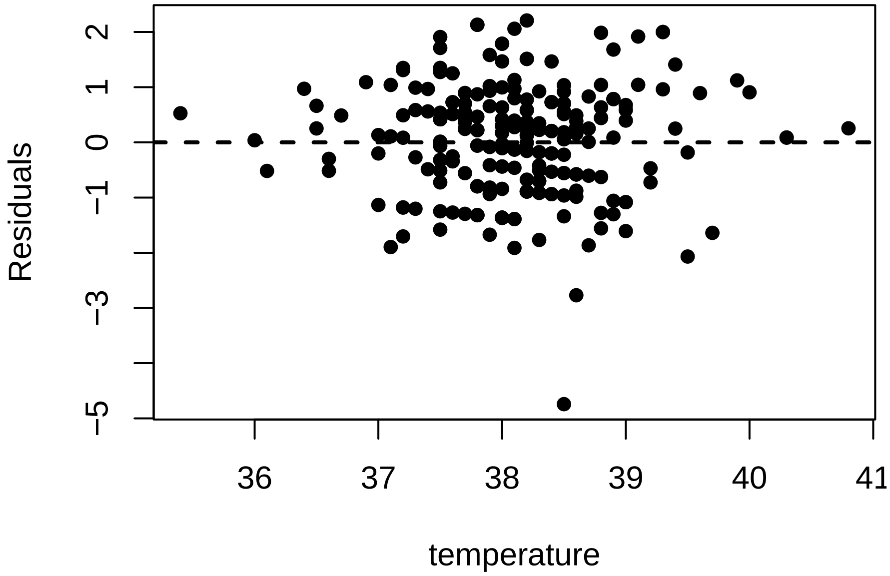
Problem 12. *Proof.*

```r
# Weighted LS using temperature
fit = lm(Y ~ temperature, horse)
summary(fit)
```

```
##
## Call:
## lm(formula = Y ~ temperature, data = horse)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.60814 -0.38480  0.07008  0.35146  1.21521
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -1.72473    1.87907  -0.918  0.35962
## temperature  0.13055    0.04929   2.649  0.00862 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.5515 on 238 degrees of freedom
## Multiple R-squared:  0.02863,    Adjusted R-squared:  0.02455
## F-statistic: 7.015 on 1 and 238 DF,  p-value: 0.008623
```

```r
X = model.matrix(fit)
n = dim(X)[1]
p = dim(X)[2] - 1
residuals = fit$residuals
```

```r
plot(rstandard(fit) ~ temperature, horse, ylab = "Standardized
Residuals",
    pch = 16)
abline(h = 0, lty = 2, lwd = 2)
```



The variance of the residuals appears to be an increasing function of temperature. Since the weights are inversely proportional to the residual variances, we infer that the weights would need to be a decreasing function of temperature.

```r
w = rep(1, n)
err = Inf
tol = 1e-10
niter = 0
maxiter = 1000
while (err > tol & niter < maxiter) {
    aux = lm(log(residuals^2) ~ log(temperature), horse)
    err = sum(abs(w - exp(-aux$fitted.values)))/sum(w)
    w = exp(-aux$fitted.values)
    WLS = lm(Y ~ temperature, horse, weights = w)
    residuals = WLS$residuals
    niter = niter + 1
}
summary(aux)
```

```
##
## Call:
## lm(formula = log(residuals^2) ~ log(temperature), data = horse)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
```

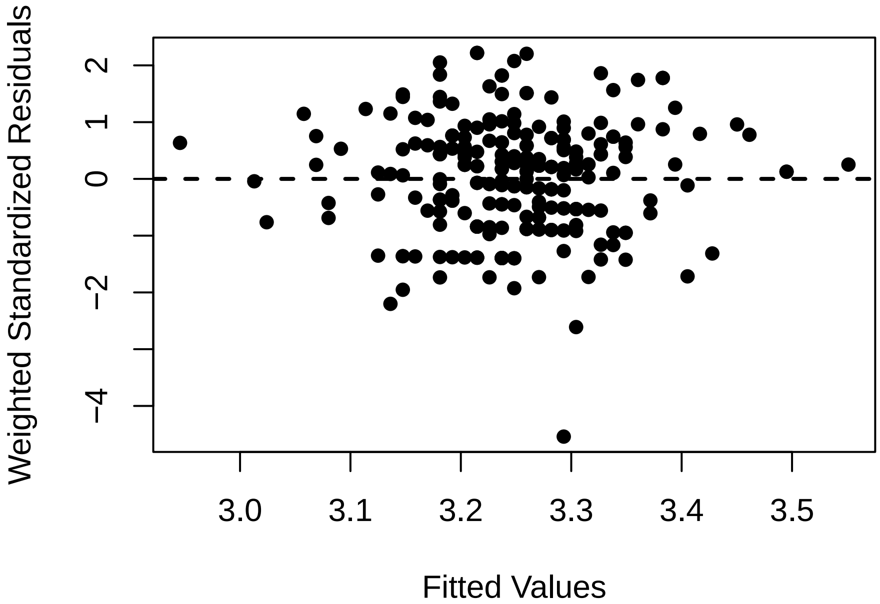
```
## -10.6695  -0.9516   0.3997   1.4790   4.2089
##
## Coefficients:
##                  Estimate Std. Error t value Pr(>|t|)
## (Intercept)       -36.344     25.195  -1.443    0.150
## log(temperature)    9.326      6.921   1.348    0.179
##
## Residual standard error: 2.032 on 238 degrees of freedom
## Multiple R-squared:  0.007572,   Adjusted R-squared:  0.003402
## F-statistic: 1.816 on 1 and 238 DF,  p-value: 0.1791
```

```r
summary(WLS)
```

```
##
## Call:
## lm(formula = Y ~ temperature, data = horse, weights = w)
##
## Weighted Residuals:
##     Min      1Q  Median      3Q     Max
## -8.2025 -1.2296  0.2316  1.1614  4.0115
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -1.02349    1.82516  -0.561   0.5755
## temperature  0.11212    0.04804   2.334   0.0204 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.812 on 238 degrees of freedom
## Multiple R-squared:  0.02238,    Adjusted R-squared:  0.01827
## F-statistic: 5.448 on 1 and 238 DF,  p-value: 0.02042
```

We estimate that $\sigma_i^2 \approx 1.64 \times 10^{-16} \cdot temperature_i^{9.33}$

```r
plot(WLS$fitted.values, rstandard(WLS), xlab = "Fitted Values", ylab = "Weighted Standar
    pch = 16)
abline(h = 0, lty = 2, lwd = 2)
```




```r
WLS_all = lm(Y ~ temperature + inv_pulse + log_pcv + log_tp, horse, weights = w)
summary(WLS_all)
```

```
##
## Call:
## lm(formula = Y ~ temperature + inv_pulse + log_pcv + log_tp,
##     data = horse, weights = w)
##
## Weighted Residuals:
##    Min     1Q Median     3Q    Max
## -7.952 -1.024 -0.029  1.025  4.173
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   0.10632    1.79804   0.059   0.9529
## temperature   0.08797    0.04452   1.976   0.0493 *
## inv_pulse   -36.80128    6.79445  -5.416  1.5e-07 ***
## log_pcv       0.12386    0.13978   0.886   0.3765
## log_tp       -0.02853    0.03124  -0.913   0.3621
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.673 on 235 degrees of freedom
## Multiple R-squared:  0.1768, Adjusted R-squared:  0.1628
## F-statistic: 12.62 on 4 and 235 DF,  p-value: 2.579e-09
```

The WLS standard errors are slightly smaller than the corresponding OLS estimate.

```r
# t-test and CIs
t_WLS = coef(WLS_all)/summary(WLS_all)$coefficients[, "Std. Error"]
```

```
t_WLS
```

```
## (Intercept) temperature   inv_pulse     log_pcv      log_tp
##  0.05913009  1.97605238 -5.41637149  0.88611701 -0.91323043
```

```
CI_WLS = confint(WLS_all, level = 0.95)
CI_WLS
```

```
##                     2.5 %       97.5 %
## (Intercept) -3.436017e+00   3.64865376
## temperature  2.645365e-04   0.17567304
## inv_pulse   -5.018710e+01 -23.41546165
## log_pcv     -1.515217e-01   0.39924636
## log_tp      -9.007966e-02   0.03301842
```

```
t_OLS = coef(quant_model)/summary(quant_model)$coefficients[, "Std. Error"]
t_OLS
```

```
## (Intercept) temperature   inv_pulse     log_pcv      log_tp
##   0.3445446   1.6332101  -5.4308577   0.8144766  -0.9818990
```

```
CI_OLS = confint(quant_model, level = 0.95)
CI_OLS
```

```
##                   2.5 %       97.5 %
## (Intercept)  -3.02626049   4.30911449
## temperature  -0.01560159   0.16686712
## inv_pulse   -51.63124263 -24.14315363
## log_pcv      -0.16066009   0.38712271
## log_tp       -0.09471723   0.03170743
```

Temperature became significant under WLS, where is was insignificant in OLS. Inv_pulse is highly significant in both. The other coefficients don't change much in significance. The temperature CI under WLS does not include 0, and it does in OLS. WLS gives slightly narrower CIs for all covariates compared to OLS.

□

Problem 13. *Proof.*

```
# mean comparisons for pain
Q = factor(horse$pain)
n = length(Q)
K = length(levels(Q))
mu = aggregate(Y ~ pain, data = horse, FUN = function(x) mean(x))[, 2]
sigma = sqrt(sum((Y - mu[Q])^2)/(n - K))
SE = sigma * sqrt(2 * K/n)
diff = outer(mu, mu, "-")
diff = diff[lower.tri(diff)]
```

```r
lab = outer(levels(Q), levels(Q), function(x, y) {
    paste0(x, "-", y)
})
names(diff) = lab[lower.tri(lab)]
CI = cbind(diff - qt(1 - alpha/2, n - K) * SE, diff + qt(1 - alpha/2,
    n - K) * SE)
colnames(CI) = c("Lower Bound", "Upper Bound")
print(CI)
```

```
##                        Lower Bound  Upper Bound
## continuous-alert        0.23287316  0.668893975
## depressed-alert         0.16919893  0.605219750
## mild-alert             -0.03972276  0.396298053
## severe-alert            0.07428923  0.510310049
## depressed-continuous   -0.28168463  0.154336183
## mild-continuous        -0.49060633 -0.054585513
## severe-continuous      -0.37659433  0.059426483
## mild-depressed         -0.42693211  0.009088712
## severe-depressed       -0.31292011  0.123100708
## severe-mild            -0.10399841  0.332022404
```

```r
# Tukey's Honestly Significant Difference Method cannot do manual
# calculations because group sizes are unequal
TukeyHSD(aov(Y ~ pain, horse))
```

```
##   Tukey multiple comparisons of means
##     95% family-wise confidence level
##
## Fit: aov(formula = Y ~ pain, data = horse)
##
## $pain
##                               diff         lwr        upr     p adj
## continuous-alert        0.45088357  0.09060383 0.81116330 0.0061313
## depressed-alert         0.38720934  0.09141242 0.68300626 0.0035506
## mild-alert              0.17828764 -0.10471823 0.46129352 0.4164565
## severe-alert            0.29229964 -0.05089540 0.63549468 0.1356510
## depressed-continuous   -0.06367423 -0.40476160 0.27741315 0.9859950
## mild-continuous        -0.27259592 -0.60265213 0.05746029 0.1582325
## severe-continuous      -0.15858393 -0.54150377 0.22433592 0.7859018
## mild-depressed         -0.20892170 -0.46705234 0.04920895 0.1741052
## severe-depressed       -0.09490970 -0.41789877 0.22807937 0.9280458
## severe-mild             0.11401200 -0.19730528 0.42532927 0.8520905
```

Tukey's Honestly Significant Difference Method (Tukey's HSD) calculates pairwise differences using a critical value from Tukey's studentized range distribution rather than the t-distribution. It can only take positive values and is parametrized by the number of groups (k) and the residual (n-k) degrees of freedom. For each pair of group means $\hat{\mu}_k - \hat{\mu}_\ell$, simultaneous 95% CIs are computed such that FWER $\leq \alpha$ using the upper α-quantile of the studentized range divided by $\sqrt{2}$ and multiplied by the pooled sample standard deviation times the square root of 2 times the number of groups divded by the sample size, eg $\hat{\mu}_k - \hat{\mu}_\ell \pm \frac{R_{K,n-K;\alpha}}{\sqrt{2}} \cdot S\sqrt{\frac{2K}{n}}$.

The CIs using the t-distribution are generally wider than the narrower Tukey's estimates. Tukey's estimates take into account the studentized range distribution, which adjusts for the fact that multiple comparisons are being made simultaneously while controlling the FWER. Since Tukey's is more conservative, there are less significant comparisons under this method. Only 2 comparisons are significant under Tukey's (continuous-alert and depressed-alert). These CIs do not contain 0. For the CIs using the t-distribution, both of those comparisons are significant, as well as two others (severe-alert and mild-continuous). □

Problem 14.

```r
# Global Null Testing
Y = log(horse$respiratory_rate)

fit <- lm(Y ~ surgery + age + temperature + pulse + inv_pulse + temperature_of_extremiti
    peripheral_pulse + pain + peristalsis + abdominal_distension + packed_cell_volume +
    total_protein + outcome + surgical_lesion, data = horse)
summary(fit)
```

```
##
## Call:
## lm(formula = Y ~ surgery + age + temperature + pulse + inv_pulse +
##     temperature_of_extremities + peripheral_pulse + pain + peristalsis +
##     abdominal_distension + packed_cell_volume + total_protein +
##     outcome + surgical_lesion, data = horse)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.16240 -0.26064  0.00095  0.33168  1.19364
##
## Coefficients:
##                                   Estimate Std. Error t value Pr(>|t|)
## (Intercept)                      6.676e-01  2.044e+00   0.327  0.74428
## surgery1                         3.772e-02  8.879e-02   0.425  0.67138
## ageyoung                         6.839e-01  1.786e-01   3.828  0.00017 ***
## temperature                      6.306e-02  4.994e-02   1.263  0.20805
## pulse                           -3.539e-03  3.833e-03  -0.923  0.35690
## inv_pulse                       -2.272e+01  1.790e+01  -1.269  0.20577
## temperature_of_extremitiescool  -1.468e-01  1.229e-01  -1.195  0.23358
## temperature_of_extremitiesnormal -1.991e-01  1.373e-01  -1.450  0.14844
## temperature_of_extremitieswarm   9.989e-02  1.559e-01   0.641  0.52249
## peripheral_pulseincreased        8.005e-01  2.769e-01   2.890  0.00425 **
## peripheral_pulsenormal           2.042e-01  1.781e-01   1.146  0.25300
```

```
## peripheral_pulsereduced            3.649e-01  1.692e-01   2.156  0.03220 *
## paincontinuous                     2.529e-01  1.535e-01   1.648  0.10088
## paindepressed                      2.024e-01  1.186e-01   1.706  0.08941 .
## painmild                           4.295e-02  1.050e-01   0.409  0.68285
## painsevere                         1.195e-01  1.384e-01   0.863  0.38910
## peristalsishypermotile             2.276e-02  1.265e-01   0.180  0.85737
## peristalsishypomotile              1.328e-01  9.421e-02   1.409  0.16022
## peristalsisnormal                  1.280e-01  1.589e-01   0.806  0.42132
## abdominal_distensionnone           1.738e-02  9.832e-02   0.177  0.85986
## abdominal_distensionsevere         1.611e-01  1.291e-01   1.248  0.21342
## abdominal_distensionslight         7.998e-02  9.971e-02   0.802  0.42338
## packed_cell_volume                 4.712e-03  4.205e-03   1.120  0.26381
## total_protein                     -6.036e-04  1.316e-03  -0.459  0.64687
## outcomeeuthanized                  2.111e-02  1.352e-01   0.156  0.87609
## outcomelived                       9.959e-02  1.039e-01   0.959  0.33870
## surgical_lesion1                   1.411e-01  9.759e-02   1.446  0.14966
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4859 on 213 degrees of freedom
## Multiple R-squared:  0.3251, Adjusted R-squared:  0.2427
## F-statistic: 3.946 on 26 and 213 DF,  p-value: 1.059e-08
```

```r
alpha = 0.05
pvals <- summary(fit)$coefficients[-1, 4]
m <- length(pvals)

sum(pvals < alpha)
```

```
## [1] 3
```

```r
# Bonferroni
Bonf = min(pvals)
print(Bonf < alpha/m)
```

```
## [1] TRUE
```

```r
# Fishers Combo
Fisher = -2 * sum(log(pvals))
print(Fisher > qchisq(1 - alpha, 2 * m))
```

```
## [1] TRUE
```

```r
# Simes
Simes = sum(pvals[order(pvals)] <= (1:m) * alpha/m)
print(Simes > 0)
```

```
## [1] TRUE
```

We tested the global null hypothesis that all regression coefficients are equal to zero against the alternative that at least one coefficient is nonzero. The overall F-test from the regression output provides a global hypothesis test that controls the family-wise error rate. The F-statistic of 60.45 with a p-value less than2.2 $\times 10^{-16}$ leads us to reject the global null hypothesis. Therefore, there is strong evidence that at least one covariate has a statistically significant effect on the response variable.

After applying both Bonferroni correction to control the family-wise error rate, the Simes Global Test, and the Benjamini–Hochberg procedure to control the false discovery rate at the 5% level, all tested effects remained statistically significant and came to the same conclusion. Bonferroni controls the probability of any false positive and is very conservative, where BH allows some false positives but increases power, and Simes tests the global null that all effects are zero.

□

Problem 15. *Proof.*

```
# Holm-Bonferroni Test
HB = numeric(m)
for (j in 1:m) {
    HB[j] = min(max((m - (1:j) + 1) * pvals[order(pvals)][1:j]), 1)
}
HB = HB[order(order(pvals))]
HB = p.adjust(pvals, "holm")
sum(HB < alpha)
```

```
## [1] 1
```

```
names(pvals)[HB < alpha]
```

```
## [1] "ageyoung"
```

```
# Benjamini-Hochberg Procedure
BH = numeric(m)
for (j in 1:m) {
    BH[j] = min(min(m * pvals[order(pvals)][j:m]/(j:m)), 1)
}
BH = BH[order(order(pvals))]
BH = p.adjust(pvals, "BH")
sum(BH < alpha)
```

```
## [1] 1
```

```
names(pvals)[BH < alpha]
```

```
## [1] "ageyoung"
```

```
# Benjamini-Yekutieli Procedure
```

```r
BY = numeric(m)
for (j in 1:m) {
    BY[j] = min(min(m * sum(1/(1:m)) * pvals[order(pvals)][j:m]/(j:m)),
        1)
}
BY = BY[order(order(pvals))]
BY = p.adjust(pvals, "BY")
sum(BY < alpha)
```

```
## [1] 1
```

```r
names(pvals)[BY < alpha]
```

```
## [1] "ageyoung"
```

All three methods yield identical conclusions in this dataset. After adjusting for multiple testing and controlling for other covariates, young horses exhibit a statistically significant difference in respiratory rate compared to the reference age group (old in this case). This was interesting to see compared to the uncorrected p-values showed 3 coefficients having raw p-values < 0.05 before controlling FWER and FDR showcasing how these controls yield more conservative conclusions.

□

Problem 16. *Proof.*

Transformations. Exploratory data analysis, residual diagnostics, and a Box–Cox profile likelihood indicated violations of normality and homoscedasticity for the raw response. The Box–Cox confidence interval included $\lambda = 0$, motivating a log transformation of the response. Several covariates exhibited skewness and leverage, and were therefore transformed using these transformations:

$$\log(\text{respiratory rate}), \quad \text{inv\_pulse} = 1/\text{pulse}, \quad \log(\text{packed cell volume}), \quad \log(\text{total protein}).$$

These transformations improved symmetry, reduced outlier influence, and stabilized variance.

Fixed effects and interactions. Fixed effects were selected using best subset selection with validation mean squared prediction error. Temperature, inverse pulse, and pain category consistently showed meaningful associations with the response. Interaction terms between pain and quantitative predictors were tested and removed when unsupported, resulting in a main-effects formula that avoids overfitting.

Random effects. To account for dependence caused by grouped categorical variables, a mixed-effects framework was considered. Random intercepts were included for variables with nonzero variance components, including age, peripheral pulse, peristalsis, temperature of extremities, and surgical lesion. Intraclass correlation coefficients indicated small to moderate clustering effects, justifying random effects to prevent underestimated standard errors and invalid inference.

Non-constant variance. Formal testing revealed heteroscedasticity across several qualitative predictors (notably pain and abdominal distension), even after transformation. To address this, inference was based on heteroscedasticity-consistent (HC3) covariance estimators, which produced wider and more reliable confidence intervals. In addition, weighted least squares was explored, with weights modeled as a decreasing function of temperature, yielding more efficient estimates while preserving qualitative conclusions.

Inference methods. Inference was carried out using multiple complementary approaches to ensure robustness:

- Asymptotic t, F, and Wald tests for baseline comparison.
- Nonparametric permutation tests for both global and individual hypotheses, avoiding reliance on distributional assumptions.
- Bootstrap confidence intervals (normal, percentile, and pivotal), which revealed greater uncertainty than OLS-based intervals and guarded against finite-sample bias.

Regularization and stability. Ridge and lasso regression were used to assess sensitivity to collinearity and overfitting. Ridge regression shrank coefficients toward zero and confirmed stability of effect estimates, while lasso did not materially alter conclusions due to weak collinearity among predictors.

Multiple testing control. Given the large number of covariates and pairwise comparisons, multiple testing corrections were applied. Global null hypotheses were assessed using overall F-tests, while family-wise error rate was controlled using Bonferroni and Holm procedures. False discovery rate was controlled using Benjamini–Hochberg and Benjamini–Yekutieli adjustments. For pairwise mean comparisons among pain categories, Tukey's honestly significant difference procedure was used to obtain simultaneous confidence intervals.

Final model. The final inferential model is a log-linear regression with transformed covariates, selected fixed effects, and appropriate random effects, with inference based on HC3 standard errors and supported by bootstrap and permutation methods.

OR

To obtain reliable inference, the final model was specified on the log scale for respiratory rate. A Box–Cox analysis supported a log transformation, which improved normality and substantially reduced heteroscedasticity. Several marginal effects that appeared significant on the original scale disappeared after transformation, indicating that inference on the log scale is more stable and reliable. Predictor transformations were also applied where appropriate: inverse pulse was used due to a stronger and more linear relationship with the response, packed cell volume and total protein were log-transformed to address skewness, and temperature was left untransformed.

Both fixed and random effects were incorporated based on variance component analysis. Age explained the largest random-effect variance and was therefore modeled as a random effect, along with peripheral pulse, peristalsis, and temperature of extremities. Predictors with near-zero variance components were treated as fixed effects. Interaction terms were explored but were not consistently significant and increased model complexity without improving inference, so they were excluded from the final model. Despite transformation, heteroscedasticity persisted for some qualitative predictors, particularly pain and abdominal distension, making classical OLS standard errors unreliable; as a result, heteroscedasticity-consistent (robust) standard errors were used. Finally, multiple testing adjustments were essential: unadjusted p-values overstated significance, and after applying Holm (FWER) and Benjamini–Hochberg/Benjamini–Yekutieli (FDR) corrections, only age remained statistically significant, underscoring the importance of correcting for multiple comparisons.